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"Department of Electronics and Communication Engineering"

REPORT ON
"PCB DESIGN WORKSHOP"

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[ANALOG LAYOUT ENGINEER, TEXAS INSTRUMENTS]

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1. INTRODUCTION

Printed Circuit Board (PCB) designing is a crucial process in electronics that involves creating the physical layout of electronic circuits on a flat insulating board using conductive copper tracks. These copper tracks form electrical connections between various electronic components such as resistors, capacitors, diodes, transistors, and integrated circuits. The PCB provides mechanical support to components and ensures proper electrical connectivity, replacing traditional wired connections and making the circuit more compact, organized, and reliable. The design process begins with schematic creation, where the logical connections of the circuit are defined using PCB design software, followed by component placement and routing of copper traces to establish electrical paths.

PCB designing plays an important role in improving circuit performance, reducing noise and interference, and enabling efficient power and signal transmission. It allows the development of complex electronic systems in a small and efficient form factor, making it essential in modern electronic devices. PCB design is widely used in applications such as computers, embedded systems, FPGA boards, communication systems, medical equipment, aerospace, and consumer electronics. Software tools like KiCad, Altium Designer, Eagle, and OrCAD are commonly used to design, simulate, and verify PCB layouts before fabrication, ensuring accuracy, reliability, and manufacturability of the electronic circuit.

In addition, PCB designing involves selecting appropriate board materials, defining the number of layers, and ensuring proper spacing and layout to meet electrical and manufacturing requirements. Designers must follow design rules such as trace width, clearance, and via placement to ensure reliable operation and prevent issues like short circuits, signal loss, and overheating. Advanced PCB designs may include multilayer boards to support complex circuits and high-speed applications. Proper PCB design also improves thermal management, enhances durability, and supports efficient mass production. As modern electronic systems continue to become smaller and more powerful, PCB designing has become an essential skill for engineers working in electronics, embedded systems, FPGA, and hardware development.

2. OBJECTIVES

- Understanding the fundamental concepts and significance of Printed Circuit Board design in electronic systems.
- Learning schematic creation, component placement, and routing techniques using PCB design software.
- Gaining practical knowledge of PCB layout elements such as tracks, pads, vias, layers, and footprints.
- Understanding PCB design flow, design rules, and methods to ensure reliable and manufacturable PCB layouts.
- Developing hands-on skills in PCB designing and awareness of its applications in embedded systems, FPGA, and modern electronic devices.

3. DAY WISE SESSIONS

3.1 DAY-1: Basics of PCB and Electronic Components

The first day of the PCB Design workshop focused on introducing the fundamental concepts of Printed Circuit Boards and their importance in modern electronic systems. A Printed Circuit Board (PCB) is a flat insulating board used to mechanically support and electrically connect electronic components using conductive copper tracks. The session began with an explanation of the role of PCBs in various applications such as computers, mobile phones, embedded systems, medical devices, and aerospace electronics. The advantages of PCBs over traditional wiring methods, including compact size, improved reliability, reduced noise, and ease of mass production, were also discussed.

The session continued with an introduction to basic electronic components such as resistors, capacitors, LEDs, switches, and integrated circuits. The functions and working principles of each component were explained in detail. Resistors were described as components that limit current flow, capacitors as energy storage devices, LEDs as light-emitting indicators, and integrated circuits as complex devices that perform various electronic functions. Participants also learned about circuit symbols and schematic diagrams, which represent electronic circuits in a graphical format.

The concept of schematic design was introduced as the first step in PCB designing. The schematic shows logical connections between components without considering their physical placement. Participants were introduced to PCB design software tools such as KiCad and EasyEDA, which are used to create schematics and PCB layouts. A live demonstration was given to show how to create a simple schematic using these tools. The importance of schematic accuracy and proper component selection was emphasized, as errors in this stage can affect the entire PCB design process.

Overall, the first day provided a strong foundation in electronics and PCB fundamentals, helping participants understand the importance of PCB design and preparing them for the practical design and fabrication sessions in the following days.

3.2 DAY-2: PCB Layout Design and Design Rules

The second day of the workshop focused on PCB layout design, which involves converting the schematic diagram into a physical layout on the PCB. This process is essential for transforming the logical circuit into a manufacturable board. The structure of a PCB was explained in detail, including elements such as the top layer, bottom layer, copper traces, pads, and vias. Copper traces act as electrical connections, pads provide locations for soldering components, and vias connect different layers of the PCB. Understanding these elements helped participants learn how a PCB forms the foundation of an electronic circuit and ensures proper signal and power flow.

Component placement was another important topic covered during the session. Proper placement of components is necessary to ensure efficient routing, minimize signal interference, and improve overall circuit performance. Participants learned how to arrange components logically based on their function and connectivity. Placing related components closer reduces trace length and improves signal integrity. The importance of proper grounding and power routing was also explained, as incorrect placement can lead to noise, signal distortion, and circuit malfunction. Good placement also makes the PCB easier to assemble and maintain.

The session also emphasized the importance of following PCB design rules to ensure reliability and manufacturability. Participants learned about critical parameters such as trace width, spacing between traces, pad sizes, and via dimensions. These design rules help prevent issues such as short circuits, overheating, and signal loss. The Design Rule Check (DRC) feature in PCB design software was introduced, which helps identify errors and ensures the design meets manufacturing standards. Following these rules ensures that the PCB functions correctly and can be fabricated without defects.

Hands-on practice was provided using PCB design software such as KiCad or EasyEDA. Participants learned how to import schematics, assign footprints to components, place components on the board, and route copper traces to create complete PCB layouts. They also learned how to verify their designs and correct errors using software tools. This practical experience helped participants understand the complete PCB layout design process and its importance in developing reliable and efficient electronic circuits.

3.3 DAY-3: PCB Fabrication Process

The third day of the workshop focused on the PCB fabrication process, which involves converting the designed PCB layout into a physical circuit board. Participants were introduced to the toner transfer method, which is widely used for prototyping PCBs. In this method, the PCB layout created using design software is printed onto glossy paper using a laser printer. The printed layout is then transferred onto the copper-clad board using heat and pressure with the help of an iron box. The heat causes the toner to stick to the copper surface, forming a protective layer that defines the circuit pattern. This process helped participants understand how digital PCB designs are physically implemented on copper boards.

Before transferring the design, proper preparation of the copper-clad board was explained and demonstrated. The copper surface was cleaned thoroughly using abrasive material or cleaning solution to remove dust, grease, and oxidation. This cleaning process is essential to ensure proper adhesion of the toner to the copper surface and to avoid defects in the circuit traces. Participants learned that improper cleaning can result in broken or incomplete traces, which can affect circuit functionality. After cleaning, the printed layout was carefully aligned and transferred onto the copper board using controlled heat.

After successfully transferring the layout, the etching process was carried out using Ferric Chloride solution. Etching is the process of removing unwanted copper from the board, leaving only the required circuit traces protected by the toner. Participants observed how the exposed copper dissolved in the chemical solution, revealing the designed circuit pattern. Safety precautions such as wearing gloves, avoiding direct contact with chemicals, and working in a safe environment were emphasized during this process. After etching, the board was washed with water to remove chemical residues and cleaned to reveal the final copper traces clearly.

This session provided valuable practical exposure to the PCB fabrication process and helped participants understand how theoretical PCB designs are transformed into real circuit boards. The importance of precision, careful handling, and following proper procedures was emphasized to ensure successful PCB fabrication. Participants gained hands-on experience in toner transfer and etching, which are essential skills in PCB prototyping and hardware development.

3.4 DAY-4: Drilling and Soldering Techniques

The fourth day of the workshop focused on drilling and soldering, which are essential processes for assembling electronic components onto the fabricated PCB. Participants were introduced to PCB drilling techniques used to create holes for through-hole components such as resistors, LEDs, and integrated circuits. A PCB drilling machine was used to drill precise holes at designated pad locations. The importance of accurate drilling was emphasized, as improper drilling can damage copper traces and affect circuit functionality. Safety precautions, including proper handling of drilling tools and maintaining steady positioning of the PCB, were explained to ensure both user safety and board quality.

The soldering process was explained in detail as a critical step for establishing both electrical and mechanical connections between components and the PCB. Participants were introduced to the soldering iron, solder wire, and supporting tools used in the soldering process. The correct technique of heating both the component lead and the PCB pad simultaneously before applying solder was demonstrated. This ensures proper melting and bonding of the solder, resulting in strong and reliable connections. The importance of maintaining the correct temperature and avoiding excessive heating was also discussed to prevent damage to components and PCB pads.

Common soldering defects were also explained to help participants identify and avoid errors. These defects included cold solder joints, which occur due to insufficient heating, excess solder that can cause short circuits, and weak joints that may result in unreliable connections. Participants learned how to visually inspect solder joints and ensure they are smooth, shiny, and properly formed. Proper soldering practices help improve circuit reliability, performance, and long-term durability.

Hands-on practice was provided, allowing participants to solder electronic components such as resistors, LEDs, and ICs onto their fabricated PCBs. This practical session helped participants gain confidence and improve their technical skills in assembling electronic circuits. The session emphasized the importance of precision, patience, and proper technique in soldering to ensure successful circuit assembly and reliable operation of electronic devices.

3.5 DAY-5: Circuit Testing and Debugging

The final day of the workshop focused on testing and debugging the assembled PCB circuits to ensure proper functionality and reliability. Participants were introduced to the use of a multimeter, which is an essential tool for measuring electrical parameters and verifying circuit connections. The concept of continuity testing was explained, which helps determine whether electrical connections between points are complete. Participants also learned how to measure voltage across different components and identify short circuits that could affect circuit performance. This session helped participants understand the importance of testing in ensuring the correctness of PCB circuits.

The session also covered troubleshooting techniques used to identify and correct errors in PCB circuits. Participants learned how to inspect the PCB visually to detect issues such as broken copper traces, poor solder joints, and incorrect component placement. The importance of checking component orientation, especially for polarized components such as LEDs and integrated circuits, was emphasized. Participants were taught systematic debugging methods to locate faults and correct them effectively. This process helped them understand how errors in fabrication or assembly can affect circuit operation and how they can be resolved.

After completing the testing and debugging process, participants powered their circuits using an appropriate power source to verify their functionality. The working of the circuit confirmed that the PCB design, fabrication, drilling, and soldering processes were performed correctly. Participants observed the successful operation of their circuits, which demonstrated the complete implementation of their PCB from design to final output. This step helped reinforce their understanding of the entire PCB development cycle.

The workshop concluded with a summary of all the concepts and practical skills learned throughout the sessions. Participants gained comprehensive knowledge of PCB design, layout, fabrication, drilling, soldering, and testing processes. The workshop provided valuable hands-on experience and helped participants develop practical skills required for electronics and hardware development. This experience prepared participants for future projects and careers in electronics, embedded systems, and PCB design.

4. CONCLUSION & FEEDBACK

Conclusion

The PCB Design Workshop provided comprehensive knowledge and practical experience in the complete process of designing and fabricating Printed Circuit Boards. The workshop began with the fundamentals of electronic components and schematic design, followed by PCB layout creation using design software. Participants learned important concepts such as component placement, routing, design rules, and the use of PCB design tools. These sessions helped build a strong theoretical foundation in PCB design and its role in modern electronic systems.

The workshop also provided hands-on experience in PCB fabrication using the toner transfer and etching process. Participants learned how to prepare copper-clad boards, transfer circuit layouts, and safely perform etching to obtain the required circuit traces. Further sessions focused on drilling and soldering techniques, where participants assembled electronic components onto their fabricated PCBs. Proper soldering methods and safety precautions were emphasized to ensure reliable electrical and mechanical connections.

The final sessions focused on testing and debugging the assembled circuits using a multimeter. Participants learned how to verify circuit continuity, identify faults, and ensure proper functionality of their PCB circuits. This helped them understand the importance of testing and quality verification in electronics manufacturing.

Overall, the workshop successfully bridged the gap between theoretical knowledge and practical implementation of PCB design. It enhanced participants' technical skills, practical understanding, and confidence in designing and developing electronic circuits. The knowledge and hands-on experience gained from this workshop will be valuable for future academic projects and careers in electronics, embedded systems, and hardware design.

Feedback

The PCB Design Workshop was a highly informative and valuable learning experience that provided both theoretical knowledge and practical exposure to PCB design and fabrication. The workshop was well organized, and the instructors explained each concept clearly, making it easy to understand the complete PCB development process. The hands-on sessions, including PCB layout design, fabrication using toner transfer and etching, drilling, soldering, and testing, helped in developing practical skills and confidence in assembling electronic circuits.

5.PICTURES

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